

A Rule-Constrained Large Language Model Pipeline for Patient-Friendly Interpretation of Vietnamese Blood Test Results

Anonymous Author(s)

Anonymous Institution

Abstract. Patients in Vietnam routinely receive laboratory reports written in clinical terminology they cannot interpret, and increasingly turn to general-purpose language models for explanations. We show this is unsafe: an unconstrained model asked to explain synthetic complete blood count panels misstated the direction of an analyte in 91.3% of cases and told patients with abnormal results that everything was normal in 32.0% of cases. We propose a four-layer pipeline in which all clinical judgement is made by deterministic rules before the language model is invoked; the model only paraphrases pre-assigned labels, and a post-generation validator rejects outputs that contradict them. On 150 synthetic panels, the full pipeline produced no unsafe output, with 64.0% generated by the model and 36.0% falling back to fixed templates. The system runs on CPU without a GPU.

Keywords: health literacy, large language models, laboratory results, patient communication, AI safety

1 Introduction

A patient leaving a Vietnamese laboratory receives a printed panel of numbers annotated with terms such as PDW, MCHC, and Monocyte, in units of G/L, fL, and pg. The reference interval is printed beside each value, so a patient can see that a number falls outside it, but not what the analyte measures, whether the deviation matters, or what to do next. The gap between possessing a result and understanding it is what health literacy describes: the ability to access, understand, appraise, and use health information to maintain one’s health.

That patients are left to close this gap alone is documented: in a study of 93 patients receiving results through an online portal, nearly two-thirds received no explanation of their results and 46% searched online to understand them [6]. Increasingly, that search means photographing the report and asking a general-purpose language model — a failure mode easy to reproduce: asked in Vietnamese what white blood cells are, the 3B-parameter model used in this study replied that they are lymphocytes (one type of white blood cell, not a synonym) and, unprompted, mentioned cancer and immune deficiency, handing an ordinary result a definitional error and two alarming diseases in one short answer.

The more consequential failure runs in the opposite direction. A model that summarises an abnormal panel as normal does not merely misinform; it removes the reason to seek care. Terminological errors are visible and invite verification, whereas false reassurance is fluent, comfortable, and acts by suppressing further action, so we treat it as a distinct error class and measure it separately: an unconstrained model produced it on 32.0% of abnormal panels in our data.

Our position is that the task should not be given to the model in the form it usually is. Deciding whether a value is abnormal is arithmetic against a reference interval, which a program does exactly; rendering that determination in plain language is what a language model does well. Separating the two, and verifying the second against the first, removes the failure mode rather than instructing it away.

This paper contributes:

1. A four-layer pipeline in which all clinical judgement is completed by deterministic rules before generation, so the model paraphrases pre-assigned labels rather than deriving them.
2. An automated post-generation validator that checks generated text against those labels and falls back to fixed templates when it does not agree, so that no unverified text reaches the reader.
3. An evaluation on 150 synthetic complete blood count panels measuring six error classes, including false reassurance, which to our knowledge has not been reported separately in prior work on patient-facing result explanation.
4. An observation that unit heterogeneity between Vietnamese laboratories introduces severe misclassification inside the rule layer itself unless explicitly normalised.

2 Related Work

2.1 Language Models for Patient-Facing Result Explanation

The closest prior work evaluates whether general-purpose models can answer patients’ questions about laboratory results. He et al. [7] compared model responses against peer answers on 53 forum question–answer pairs, assessing relevance, accuracy, helpfulness, and harmfulness, with GPT-4 outperforming smaller open models; related radiology work [2] measured readability, understandability, and anxiety-inducing potential across several models.

Assessments from within laboratory medicine are more cautious. The European Federation of Clinical Chemistry and Laboratory Medicine working group on artificial intelligence reported that such models may give superficial or incorrect answers to questions about test results and concluded they cannot be used for diagnosis [3]. On 30 interpretation questions posed by physicians, GPT-4 answered 46.7% correctly, gave partially correct answers to 23.3%, and answered 30% incorrectly or irrelevantly [8]. These evaluations concern clinicians as users; the patient-facing case is more exposed, since the reader has less basis for recognising an incorrect answer.

Two features of this literature motivate our approach. Evaluation is oriented towards readability and general accuracy; harmfulness, where assessed, is one dimension among several rather than decomposed into specific failure modes, and we are not aware of prior work that isolates false reassurance despite its distinctive consequence of suppressing care-seeking. Moreover, these studies evaluate models directly, without a mechanism to stop an unsafe response reaching the reader. Our contribution is architectural rather than evaluative: not how well a model performs the task, but how to arrange the system so its failures do not propagate.

2.2 Health Literacy in Vietnam

The premise that patients cannot interpret their own laboratory reports is well supported in the Vietnamese context. A cross-sectional study of 204 older adults in Da Nang found 60.3% with inadequate general health literacy, with urban residents scoring 2.4 times higher than rural ones and educational attainment a strong determinant [12]. Earlier work reported similar patterns [1, 4]. A recent national survey reports a combination directly relevant here: high eHealth literacy alongside low general health literacy among digitally connected adults, with online health-seeking behaviour common [10]. Patients are therefore willing to consult online tools while less equipped to evaluate them — raising, not lowering, the safety requirement on such tools.

2.3 Constraining and Verifying Model Output

Grammar-constrained decoding restricts the token distribution at each step so that only strings conforming to a grammar can be produced [5, 13], and schema-constrained generation is now a standard interface for machine-readable output.

Structural validity, however, does not entail semantic correctness: a response may satisfy a schema exactly while containing values that are wrong for the input. Our Layer 4 operates at this second level. It does not check that the output parses — the schema constraint already ensures that — but that its content agrees with labels computed independently of the model. This is closer to verification against an external source of truth than to constrained decoding, and the two are complementary: we apply schema constraints at generation time and semantic verification afterwards.

2.4 Laboratory Data and Unit Standardisation

The unit inconsistency we encountered is known at larger scale. Between 6% and 19% of laboratory tests cannot be accurately mapped to LOINC, with missing or incorrect units among the recurring obstacles [9], and an analysis of over 163 million LOINC-mapped results identified 2,019 distinct unit expressions, reduced by harmonisation to roughly 40 reference units [14]. Automated mapping approaches exist for codes [11].

This literature treats unit heterogeneity as an obstacle to data aggregation and secondary analysis. Our observation is that it is also a patient-safety issue: within our pipeline, comparing an unconverted value against a reference interval in a different unit produced a severe misclassification of a normal result, inside the deterministic layer the architecture relies on for correctness. The argument for standardisation is usually made in terms of research reuse; we offer a small piece of evidence that it also bears on what a patient is told about their own results.

3 System Design

The central design principle is that *no clinical judgement is delegated to the language model*. Whether an analyte is abnormal, and how severely, is decided by deterministic rules before the model is invoked. The model receives values that have already been labelled and is asked only to paraphrase them. A post-generation validator then checks that the generated text is consistent with those labels, and the system falls back to fixed templates when it is not.

Figure 1 shows the four layers.

3.1 Layer 1: Name and Unit Normalisation

Vietnamese laboratories name the same analyte inconsistently: a white blood cell count may appear as **WBC**, the Vietnamese abbreviation **SLBC**, the full Vietnamese phrase, or *** WBC**. Percentage and absolute counts are distinguished only by a **%** or **#** marker, which may precede or follow the name (**NEUT%** versus **% Neu**). Layer 1 resolves these through an alias dictionary; a naive normalisation stripping punctuation collapses **NEUT%** and **NEUT#** into one key, so the markers are preserved as explicit tokens.

Units differ more consequentially. Haemoglobin is reported in **g/L** by some laboratories and **g/dL** by others, a factor of ten apart. Comparing a **g/dL** value against a **g/L** reference interval places a normal result far outside range, producing a severe false alarm — and doing so inside the rule layer that the architecture relies on for correctness. Layer 1 therefore converts every value to a canonical unit, and *refuses* to process an analyte when the unit is unrecognised, or when the unit is absent and the analyte admits units of differing magnitude. Refused analytes are reported separately and excluded from the panel-level label; the system never guesses.

3.2 Layer 2: Rule-Based Classification

Each normalised value is compared against its reference interval and assigned one of three labels: *normal* (within interval), *monitor* (outside interval but within a configurable severity threshold), or *seek care* (beyond that threshold). The panel-level label is the most severe analyte label.

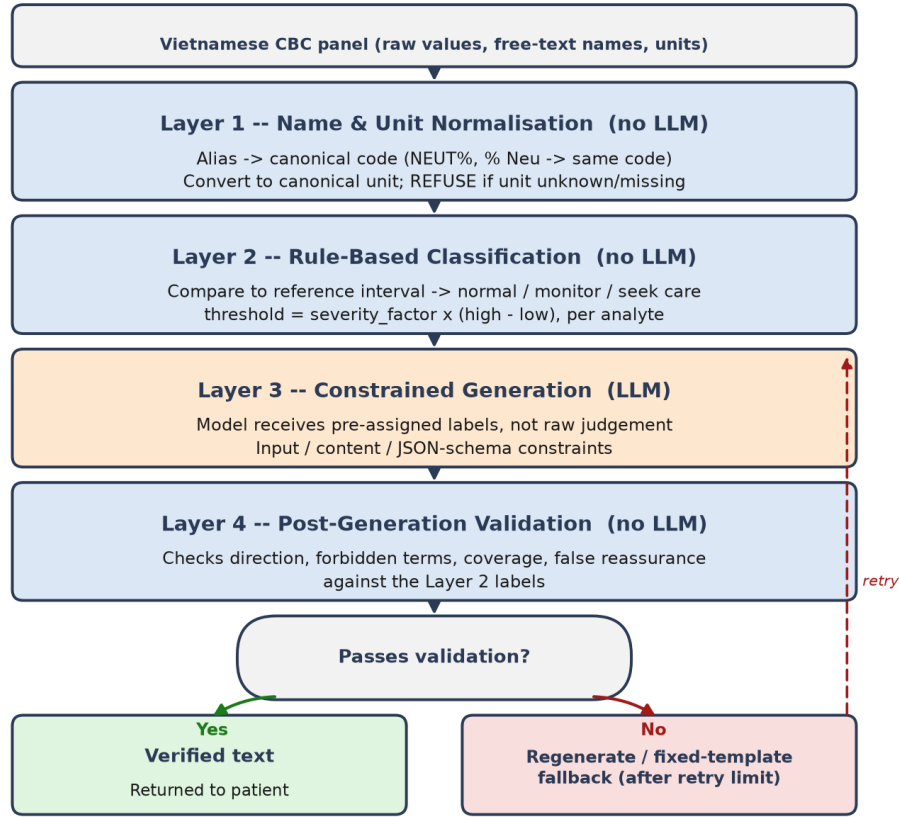


Fig. 1. Four-layer pipeline. Clinical judgement (Layer 2) precedes generation (Layer 3); Layer 4 verifies the output against Layer 2’s labels.

This layer contains no learned component. Classification accuracy is therefore a property of the design rather than an experimental result, and the language model has no opportunity to alter it.

The threshold separating *monitor* from *seek care* is defined per analyte as $severity_factor \times (high - low)$, where *high* and *low* are the bounds of the reference interval and *severity_factor* ranges from 0.15 to 0.50 across the 22 analytes used in this study. These coefficients were set heuristically for this study rather than derived from clinical outcome data or a laboratory’s critical value list; Section 7 discusses this choice.

3.3 Layer 3: Constrained Generation

The prompt supplied to the model contains, for each analyte, the value, the reference interval, the *already determined* direction and severity label, and a plain-language description of the analyte’s function. Three constraints are applied. The input constraint withholds any opportunity for independent assessment: the

model is given labels, not asked to derive them. The content constraint forbids naming diseases, attributing causes, and mentioning medications, dosages, or treatments. The output constraint requires a fixed JSON schema, which makes automated checking possible.

We use Qwen2.5-3B served locally through Ollama: laboratory data is sensitive, and a system for Vietnamese hospitals should not transmit it to external services (Section 6 returns to this).

3.4 Layer 4: Post-Generation Validation

The validator checks each generated item against the Layer 2 labels: whether the stated direction matches, whether forbidden terms appear, whether every input analyte is covered, and whether any analyte not in the input has been introduced. It additionally checks the closing statement for *false reassurance* — an assertion that all results are normal on a panel that is not.

Outputs failing validation are regenerated, up to a limit, after which the system emits a fixed template stating the value, the reference interval, and the assigned label. The pipeline therefore never returns unverified text.

Before pattern matching, generated text is normalised by stripping Vietnamese diacritics and lower-casing, and phrases that name the *reference interval* rather than assert a state — e.g. “higher than normal” (*cao hơn mức bình thường*) — are removed when they follow a comparator word, so that the word “normal” inside them is not mistaken for a status claim.

4 Evaluation

4.1 Dataset

We evaluate on 150 synthetic complete blood count panels rather than clinical records: synthetic data provides known ground-truth labels, which real reports lack, permits a balanced design across severity classes — real panels are predominantly normal and would yield few abnormal cases — and avoids handling patient data. Panels contain 10–18 analytes each, divided equally between normal, monitor, and seek-care classes, drawn using randomly selected aliases and units so that Layer 1 is exercised. Rounding is applied as laboratories report it, and any panel whose rounded values no longer match the intended class is discarded and regenerated.

4.2 Configurations

baseline Raw values and reference intervals are given to the model, which is asked to explain them for a patient. No labels, no constraints, no validation.

This approximates current practice when a patient pastes a report into a chatbot.

constrained Layers 1–3: the model receives pre-assigned labels under the three prompt constraints, but output is not verified.

full Layers 1–4, including validation, regeneration, and template fallback.

4.3 Metrics

All metrics are computed per panel by the Layer 4 validator. *Safe output* is the proportion of panels whose final output passes validation, including those served by fallback. *Model-generated pass* counts only panels the model itself produced acceptably. The remaining metrics count panels containing at least one instance of false reassurance, direction mismatch, contradictory wording, forbidden terminology, fabricated analytes, and omitted analytes respectively. The first two of those are distinct checks: *direction mismatch* compares the structured status field against the Layer 2 label, while *contradictory wording* scans the prose for wording asserting a different state, which a model can produce even when the status field is correct. All six gate whether an output passes validation.

5 Results

Table 1. Comparison across configurations, 150 synthetic panels (%). Lower is better for all error columns.

Configuration	Safe output	Model pass	False reassur.	Direction mismatch	Contra. wording	Forbid. terms	Fabric. analytes	Omitted analytes
BASELINE	0.0	0.0	32.0	91.3	56.0	79.3	32.0	71.3
CONSTRAINED	48.0	48.0	8.0	12.7	5.3	15.3	10.0	28.7
FULL	100.0	64.0	0.0	0.0	0.0	0.0	0.0	0.0

Table 1 reports the comparison; every error metric decreases monotonically across the three configurations.

Unconstrained generation is unsafe. The BASELINE configuration produced no acceptable output on any of the 150 panels. It misstated the direction of at least one analyte in 91.3% of panels and used forbidden clinical terminology in 79.3%. Most consequentially, on panels containing abnormal results it asserted that all results were normal in 32.0% of cases.

Prompt constraints help but are insufficient. Supplying pre-assigned labels reduced direction mismatch from 91.3% to 12.7% and false reassurance from 32.0% to 8.0%. Nonetheless 52.0% of CONSTRAINED outputs still failed validation: instruction alone does not guarantee compliance, so verification must be mechanical.

Validation eliminates unsafe output. The FULL configuration passed validation on all 150 panels. This figure requires qualification: 64.0% of panels were served by model-generated text, while 36.0% fell back to fixed templates after regeneration failed. The guarantee is a property of the pipeline, not evidence that a 3B model reliably performs this task, though the 64.0% model-generated rate does exceed CONSTRAINED’s 48.0%, showing regeneration recovers many initially failed outputs.

Formatting and cost. In BASELINE, 97.3% of outputs identified analytes by display name rather than canonical code and 2.0% were unparseable; both fell to zero once the schema constraint was applied. We count these separately from content errors, since it was never instructed to use that format. Mean processing time was 38.9 s for BASELINE, 35.8 s for CONSTRAINED, and 68.3 s for FULL, with a mean of 1.97 generation attempts in the latter: safety costs roughly a factor of two in latency, measured on an Intel Core i5-9400F with 24 GB RAM, no GPU.

6 Discussion

6.1 Assigning Each Component the Task It Performs Reliably

The results support a division of labour rather than an improvement in prompting. The BASELINE configuration asked the model to do two things at once: judge each value and explain it. It failed at the first in 91.3% of panels while producing fluent prose throughout. Once judgement moved to a rule layer, classification became exact by construction, and the model was left with paraphrasing, which it performs adequately.

A useful consequence is that the two components improve independently. Replacing the heuristic severity thresholds with a laboratory’s critical value list would change the labels received but nothing else in the architecture, and Table 1 would remain valid, since it measures agreement with assigned labels rather than their clinical optimality. A larger model would likewise raise the model-generated pass rate without altering Layer 4’s safety guarantee.

6.2 False Reassurance as a Distinct Metric

Evaluations of patient-facing explanation typically report readability and factual accuracy; neither captures the error we consider most damaging. A statement that all results are normal, made about a panel that is not, may score well on readability precisely because it is simple and calming — its harm lies in what it causes the reader not to do. The error is neither rare nor eliminated by instruction: 32.0% of BASELINE panels and, after explicit prompt constraints forbidding it, still 8.0% of CONSTRAINED panels, reaching zero only under mechanical verification. A prompt-level prohibition should not be accepted as evidence of its absence.

6.3 Unit Heterogeneity Is a Safety Problem, Not a Formatting One

The most instructive failure encountered during development occurred in the layer the architecture treats as trustworthy. Layer 2 is deterministic and exact — but only with respect to the numbers it receives. Presented with haemoglobin measured in g/dL and compared against a g/L interval, it classified a normal value as requiring urgent care, with complete internal consistency; no downstream verification would detect this, since every later layer would faithfully reproduce a confidently wrong label.

This reframes data standardisation from an interoperability convenience into a safety precondition: correctness for any rule-based clinical layer is conditional on canonical units, and conversion must refuse rather than guess. Coding systems such as LOINC and HL7 FHIR address exactly this; our observation is a small piece of empirical support for adopting them in Vietnamese laboratory reporting.

6.4 Deployment Considerations

Running a 3B model on CPU is slow — 68.3 s per panel under the full pipeline — but tolerable for a task performed once per report rather than interactively, and it buys two properties that matter here: patient data never leaves the institution, and no GPU is required, so the hardware barrier for a provincial hospital is a machine it already owns. The 36.0% fallback rate is the cost of a small model: for that fraction the system degrades to something no better than the original report, though no worse and never misleading.

7 Limitations

Several limitations bound these results.

No clinical validation. No medical professional reviewed the generated explanations, so outputs are shown to be internally consistent with the assigned labels, not clinically appropriate or genuinely comprehensible to patients.

Severity thresholds are not clinically grounded. The factors separating *monitor* from *seek care* were set heuristically to demonstrate the architecture, not validated against any laboratory’s critical value list, and should not be read as clinical recommendations.

Synthetic data and pattern-based validation. Panels are generated, not clinical; real reports carry noise, missing fields, and formatting variation this study does not capture. Layer 4 matches surface patterns and can be evaded by indirect phrasing, bounding the errors it detects rather than all unsafe content.

Scope. Only complete blood count is supported; biochemistry, immunology, one-sided intervals and qualitative results are not handled, and reference intervals are not stratified by sex or age. Structured input is assumed, so OCR is out of scope, and results are from a single 3B model.

8 Conclusion

We presented a pipeline for explaining blood test results to patients in which the language model is not permitted to judge whether a value is abnormal. Rules make that determination first; the model paraphrases the result; a validator checks the paraphrase against the determination and substitutes a fixed template when they disagree. Across 150 synthetic panels, false reassurance fell from 32.0% under unconstrained generation to 8.0% with prompt constraints and to zero with validation, and no unsafe output was produced by the full pipeline. The system runs on CPU without a GPU, keeping laboratory data within the institution.

9 Future Work

Several directions follow. Clinical validation is most pressing: this evaluation establishes internal consistency, not clinical appropriateness, and expert review is required before any claim of patient benefit. The heuristic severity thresholds should then be replaced with an established critical value list, coverage extended beyond the complete blood count to one-sided reference intervals and qualitative results, and a speech output layer added for elderly or low-literacy patients — since it would consume only validated text, it could introduce no content error of its own. Finally, evaluation on real multi-laboratory reports would test name and unit normalisation against variation this study could only approximate.

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